

CARGOMULE · PROBLEM STATEMENT

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DOCUMENT CONTROL

Document ID	CGM-PRB-001	Version	0.4
Status	Draft	Date	2026-09-26
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cargomule		
Source	docs/01-problem.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-26	Amish Chadha	Stronger sources
0.3	2026-09-25	Amish Chadha	TRL 3. Pitch wording "most bicycles" and \$1,000 budget per CGM-DDR-001; open questions linked to the decision record
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (users, context, constraints, out of scope, prior work with sources)
0.1	2026-09-24	Amish Chadha	Initial scaffold

CargoMule problem statement

Small businesses and households need to move 100 to 150 kg loads a few kilometers at a time, but a mainstream cargo e-bike lists at about \$2,400, the few electric-assist trailers on the market cost as much or more, and an ordinary bicycle cannot pull that weight up a hill. CargoMule aims to turn the bicycle a person already owns into a 150 kg load carrier for well under the price of a cargo e-bike.

■ The problem

Short urban and suburban trips with heavy loads (a market stall's stock, tools and materials for a trade job, a week of groceries for a large household, a delivery round) are usually made by car or van. A bicycle with a cargo trailer can do many of these trips, but the rider becomes the limit:

1. **Hills.** On level ground most people can pull about 137 kg (300 lb) on a trailer at modest speed, but a 2 % grade already triples the effort and a 4 % grade needs very low gearing ([Bikes At Work](#)). A first-order estimate for CargoMule's design load (about 188 kg of loaded trailer) on an 8 % grade is about 165 N of extra pull, or about 370 W at 8 km/h on top of the rider's own climb. That is beyond most riders.
2. **Cost of the alternatives.** A mainstream electric longtail such as the RadWagon 5 lists at \$2,399 ([Rad Power Bikes](#)). Commercial electric-assist trailers exist but are priced for businesses (see Prior work).
3. **Braking.** A heavy trailer without its own brakes pushes the bicycle when it slows. Voluntary trailer standards set low limits for unbraked trailers: 60 kg in EN 15918 and 45.4 kg (100 lb) in ASTM F1975 ([Wikipedia summary of both standards](#); [EN 15918 catalog entry](#)). A 150 kg trailer therefore needs its own brakes.

CargoMule is an electric-assist cargo trailer that hitches to most bicycles with no wiring to the bike. A load cell in the drawbar measures how hard the bicycle is pulling and the trailer's hub motor pushes in proportion, so the rider feels a fraction of the trailer's load on the flat and on hills.

■ Users and context

User	Need	Context
Market trader or street vendor	Move 80 to 150 kg of stock and a stall between storage and market	Early starts, mixed roads, curbs, parking at the stall
Tradesperson (gardener, painter, bike mechanic)	Carry tools and materials to jobs within about 10 km	Daily use, loads vary through the day, hills
Household without a car	Weekly shopping, moving furniture, bulk goods	Occasional use, stored in a shed or hallway
Community group, food bank or repair café	Pick up and deliver donations; share the trailer among members	Several riders and bikes; must hitch quickly to any bike
Local workshop or maker	Build and repair the trailer from common parts	Hand tools, a welder in some cases, e-bike parts suppliers

Operating environment

- **Roads:** paved streets, bike lanes and compacted gravel paths, with curbs, drop kerbs and grades up to about 8 % on short climbs (10 % occasionally).

- **Loads:** 100 to 150 kg of payload on the deck; towing bicycle and rider about 90 to 110 kg.
- **Speed:** 10 to 20 km/h cruising; assisted speed limited to 25 km/h in line with pedelec rules.
- **Climate:** –10 to 40 °C, rain, road spray and winter salt in some regions.
- **Charging:** household mains at home or at the business; charged overnight.

■ Constraints

- Garage-buildable prototype, \$1,000 USD or less in parts (`project.yaml` ; raised from \$900 by CGM-DDR-001, D1).
- Hitches to most bicycles with no change to the bike and no wiring to it: bikes with a quick-release or thru-axle rear wheel first, with adapters for others (see CGM-REQ-001, R7).
- Assist within e-bike limits: 250 W rated power and assist to 25 km/h at most, matching the EU pedal-assist exclusion in Regulation (EU) No 168/2013 ([EUR-Lex](#)) and within the US federal low-speed electric bicycle definition of under 750 W and 20 mph ([15 U.S.C. 2085](#)). Whether a motorized trailer falls under these rules is an open question.
- The trailer must brake itself so it never pushes the bicycle hard when slowing.
- Generic, replaceable parts (e-bike hub motor, controller, disc brakes, 20 in wheels) so a local bike shop can repair it.

■ Out of scope

- A complete cargo bicycle or tricycle (see FlatTrike).
- Carrying people or children.
- Handcart or walk-behind mode (kept out of scope by CGM-DDR-001, D10).
- Designing the battery cells or BMS (a bought-in 36 V LiFePO4 pack, CGM-DDR-001 D2; a SwapCell receiver is a later fleet variant).
- Road registration or type approval.

■ Prior work

- **Carla Cargo (Germany).** A heavy-duty trailer with an optional 250 W hub motor, 150 kg payload in its original version and an overrun brake with dual discs ([New Atlas](#)). Its current eCARLA carries up to 200 kg, uses a Bafang hub motor and a 720 Wh LiFePO4 pack, assists to 25 km/h and starts at €6,490 ([Carla Cargo](#)). The early version sensed pedaling through a crank sensor fitted to the towing bike, which ties the trailer to one bike.
- **Biomega Ein.** A single-wheel electric trailer with a 250 W hub motor that detects the bike's motion from its own wheel speed and pushes in proportion; about \$1,428 planned retail ([New Atlas](#)). It is a light-load design.
- **Roland PAXTER e.** A box trailer with two 125 W, 36 V hub motors, a 180 Wh pack and a pedal sensor, sold at about €3,300 ([Notebookcheck](#)).
- **Brake-controlling couplings.** A Bruggli patent application describes a trailer coupling that senses relative movement between two housing parts to control the trailer's brakes ([WO2022223692A1](#)), and a German utility model covers a bicycle trailer coupling for an overrun brake ([DE202010010674U1](#)). These confirm that sensing at the coupling works for braking. CargoMule uses a load cell at the coupling for propulsion as well, and a freedom-to-operate check is needed before release.

- **Unpowered heavy trailers.** Commercial cargo trailers carry 14 to 140 kg ([Wikipedia](#)), and heavy-haul builders show 137 kg is a practical unassisted limit on level ground ([Bikes At Work](#)).

The gap CargoMule addresses is an open, garage-buildable design that senses pull force at the drawbar, so it needs no sensor or wiring on the towing bike, carries 150 kg with its own brakes, and costs a fraction of a commercial electric trailer.

■ Open questions

- Is a motorized bicycle trailer legal on public roads and bike lanes in the first target country, and under which category? Carla Cargo's sale in Germany suggests a path in the EU; US state rules need checking. Proposed, awaiting Amish (CGM-DDR-001, O2).
- Which users first: a market trader group, a trades cooperative, or a community food bank? Proposed, awaiting Amish (CGM-DDR-001, O1); co-design partners are to be picked per area later.
- Is 150 kg the right payload, or do most target loads fall under 100 kg, which would allow a lighter trailer?
- How many target bikes have a quick-release or thru-axle rear wheel, and what share need a nutted-axle or hub-gear adapter?

■ User research and co-design

- ☐ Identify two or three target users (trader, tradesperson, community group) and record their typical loads, routes and storage space
- ☐ Ride typical routes with a loaded unpowered trailer and log grades and speeds
- ☐ Validate payload, range and cost assumptions with users
- ☐ Revise requirements (REQ) from findings before freezing the design